

Lab 1 - Technical Communication

UTKARSH KUMAR 23114101

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1. Let $h \in \mathbf{H}$. Define a indicator random variable $X_{i,j}$ such that

$$X_{i,j} = \begin{cases} 1, & \text{if } h(k_i) = h(k_j) \\ 0, & \text{o.w.} \end{cases}$$

Let the length of chain at index $h(k_i)$ be a random variable X_i . Then we have

$$X_i = \sum_{j=0}^{n-1} X_{i,j}$$

The expected length of chain at index $h(k_i)$ is then given by

$$\begin{aligned} E_{h \in \mathbf{H}}(X_i) &= E_{h \in \mathbf{H}}\left(\sum_{j=0}^{n-1} X_{i,j}\right) \\ &= 1 + E_{h \in \mathbf{H}}\left(\sum_{j=0, j \neq i}^{n-1} X_{i,j}\right) \\ &= 1 + \sum_{j=0, j \neq i}^{n-1} 1 \cdot \Pr_{h \in \mathbf{H}}(h(k_i) = h(k_j)) + 0 \cdot \Pr_{h \in \mathbf{H}}(h(k_i) \neq h(k_j)) \\ &= 1 + \sum_{j=0, j \neq i}^{n-1} \frac{1}{m} \\ &= 1 + \frac{n-1}{m} \end{aligned}$$

Since $m = \Theta(n) \implies \alpha = O(1) \implies$ expected length of chain at index $h(k_i)$ is constant.

2. The construction of the **dot product** family of hash function is defined as follows.

- (a) Let $|U| = m^r$ for some prime m and positive integer $r \geq 2$.
(b) For a key $k \in U$, view k in base m representation. So, we have

$$k = (k_0, k_1, \dots, k_{r-1})$$

where $k_i \in \{0, 1, \dots, m-1\}$.

- (c) Let $a = (a_0, a_1, \dots, a_{r-1})$ where a_i 's are picked uniformly at random from $\{0, 1, \dots, m-1\}$.
(d) Define $h_a(k) : U \rightarrow \{0, 1, \dots, m-1\}$ such that

$$h_a(k) = \sum_{i=0}^{r-1} a_i k_i \bmod m.$$

- (e) $\mathbf{H} = \{h_a\}$

3.

Theorem 1. *The probability of a collision in the dot product family of hash functions is $\frac{1}{m}$.*

Proof. Let $x = \{x_0, x_1, \dots, x_{r-1}\}$ and $y = \{y_0, y_1, \dots, y_{r-1}\}$ be two keys such that $x \neq y$. Then they differ in at least one digit position. Without loss of generality assume that $x_0 \neq y_0$. Then we have

$$\begin{aligned}
\Pr_{h \in \mathbf{H}}(h(x) = h(y)) &= \Pr_{h \in \mathbf{H}}\left(\sum_{i=0}^{r-1} a_i x_i \bmod m = \sum_{i=0}^{r-1} a_i y_i \bmod m\right) \\
&= \Pr_{h \in \mathbf{H}}\left(\sum_{i=0}^{r-1} a_i x_i \equiv \sum_{i=0}^{r-1} a_i y_i \bmod m\right) \\
&= \Pr_{h \in \mathbf{H}}\left(a_0(x_0 - y_0) + \sum_{i=1}^{r-1} a_i(x_i - y_i) \equiv 0 \bmod m\right) \\
&= \Pr_{h \in \mathbf{H}}\left(a_0 \equiv (x_0 - y_0)^{-1} \cdot \sum_{i=1}^{r-1} a_i(x_i - y_i) \bmod m\right) \\
&= \frac{1}{m}
\end{aligned}$$

Since $x_0 \neq y_0$ and m is prime, inverse of $x_0 - y_0$ exists. □

4. In Table 1. we show how to draw a table with L^AT_EX.

Row i	Columns				
	X_0^i	X_1^i	X_2^i	X_3^i	X_4^i
2	2	1	1	2	2
3	3	3	4	4	3
4	7	8	7	7	6
5	8	8	13	14	15
6	45	45	32	31	29

Table 1: Demo of the table.